

Example of proof in $\mathcal{H}$

Example (Proof of $A \rightarrow A$ in $\mathcal{H}$)

1. $A1$ $A \rightarrow ((A \rightarrow A) \rightarrow A)$
2. $A2$ $(A \rightarrow ((A \rightarrow A) \rightarrow A)) \rightarrow ((A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A))$
3. $MP(1, 2)$ $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$
4. $A1$ $(A \rightarrow (A \rightarrow A))$
5. $MP(4, 3)$ $A \rightarrow A$

Example of inference in $\mathcal{H}$

Example (Inference of B from $\{A, \neg A\}$ in $\mathcal{H}$)

We show that $\{A, \neg A\} \vdash_{\mathcal{H}} B$.

1. *hypothesis* A
2. $A1$ $A \rightarrow (\neg B \rightarrow A)$
3. $MP(1, 2)$ $\neg B \rightarrow A$
4. *hypothesis* $\neg A$
5. $A1$ $\neg A \rightarrow (\neg B \rightarrow \neg A)$
6. $MP(4, 5)$ $\neg B \rightarrow \neg A$
7. $A3$ $(\neg B \rightarrow \neg A) \rightarrow ((\neg B \rightarrow A) \rightarrow B)$
8. $MP(6, 7)$ $(\neg B \rightarrow A) \rightarrow B$
9. $MP(3, 8)$ B

Exercise on $\mathcal{H}$

Exercise on the Hilbert proof system

Prove that a set of formulas Γ is $\mathcal{H}$ -inconsistent if and only if $\Gamma \vdash_{\mathcal{H}} A$ for all formulas A .

Properties of $\mathcal{H}$

It is interesting to check whether the following property hold for $\vdash_{\mathcal{H}}$:

- **Reflexivity** (If $A \in \Gamma$, then $\Gamma \vdash_{\mathcal{H}} A$)
- **Ex falso sequitur quodlibet** (If Γ is unsatisfiable, then $\Gamma \vdash_{\Sigma} A$ for all A)
- **Monotonicity** (If $\Gamma \vdash_{\mathcal{H}} A$ then $\Gamma \cup \Delta \vdash_{\mathcal{H}} A$)
- **Cut** (If $\Gamma \vdash_{\mathcal{H}} A$ and $\Delta \cup \{A\} \vdash_{\mathcal{H}} B$ then $\Gamma \cup \Delta \vdash_{\mathcal{H}} B$)
- **Compactness** (If $\Gamma \vdash_{\mathcal{H}} A$, then there is a finite subset $\Gamma_0 \subseteq \Gamma$, such that $\Gamma_0 \vdash_{\mathcal{H}} A$)
- **Deduction principle** ($\Gamma \cup \{A\} \vdash_{\mathcal{H}} B$ if and only if $\Gamma \vdash_{\mathcal{H}} A \rightarrow B$)
- **Refutation principle** ($\Gamma \vdash_{\mathcal{H}} A$ if and only if $\Gamma \cup \{\neg A\}$ is $\mathcal{H}$ -inconsistent)
- **Consistency**: $\vdash_{\mathcal{H}}$ is consistent (i.e., there is no formula A such that both $\vdash_{\mathcal{H}} A$ and $\vdash_{\mathcal{H}} \neg A$)

Exercises on $\mathcal{H}$

Exercise: deduction theorem in the Hilbert proof system

Prove that the deduction theorem holds for the Hilbert proof system.

Exercise: cut principle in the Hilbert proof system

Prove that the cut principle holds for the Hilbert proof system.

Difference between logical implication and inference

- Logical implication

- ▶ **Definition:** a formula A is **logically implied** by a set of formulas Γ if every model of Γ is also a model of A
- ▶ **Notation:** $\Gamma \models A$
- ▶ **Nature:** based on semantics, it characterizes the **model theoretic view of logic**

- Inference

- ▶ **Definition:** a formula A is **inferred** by a set of formulas Γ in the proof system Σ if there is a proof of A in Σ from Γ .
- ▶ **Notation:** $\Gamma \vdash_{\Sigma} A$, or simply $\Gamma \vdash A$
- ▶ **Nature:** based on syntax (Σ is concerned with the **form** of axioms and inference rules), it characterizes the **proof theoretic view of logic**

Which is the relationship between the two?

Soundness and completeness of a proof system

Two very important properties of a proof system derive from the relationship between $\vdash_{\Sigma}$ and $\models$

Soundness

A proof system Σ is **sound** if for every formula A and every set of formula Γ , we have that $\Gamma \vdash_{\Sigma} A$ implies $\Gamma \models A$.

Completeness

A proof system Σ is **complete** if for every formula A and every set of formula Γ , we have that $\Gamma \models A$ implies $\Gamma \vdash_{\Sigma} A$.

In other words,

- a sound Σ derives only logically implied formulas, and
- a complete Σ derives all logically implied formulas.

The Hilbert proof system is sound and complete

Theorem (Soundness of Hilbert proof system)

If $\Gamma \vdash_{\mathcal{H}} A$ then $\Gamma \models A$.

Theorem (Completeness of Hilbert proof system)

If $\Gamma \models A$ then $\Gamma \vdash_{\mathcal{H}} A$.

Exercise

Prove that $\mathcal{H}$ is a sound proof system.

Completeness is much more difficult to prove ...

Importance of the Hilbert proof system

The main objective of Hilbert was to find the smallest set of axioms and inference rules from which it was possible to derive all the tautologies.

However, proofs and inferences in the Hilbert proof system are not very intuitive. Other proof systems are more intuitive (see later). As a matter of facts, nobody is practically using Hilbert system for deduction.

Why is it so important then?

The Hilbert proof systems was the **first** formal proof system formally defined and studied for any logic. At that time, proof systems were the only hope for mechanizing reasoning!

INTRODUCTION TO THE SITUATION CALCULUS

The Situation Calculus (McCarthy)

- A first order language for representing dynamically changing worlds; all changes are the result of named *actions*.
- A possible world history, which is simply a sequence of actions, is represented by a first order term called a *situation*.
- S_0 denotes the initial situation, where no actions have yet occurred.
- $do(\alpha, s)$ denotes the successor situation to s resulting from performing the action α .
- Actions may be parameterized:
 $put(x, y)$ might stand for the action of putting object x on object y ; $do(put(A, B), s)$ denotes that situation resulting from placing A on B when the world is in situation s .
- Actions are denoted by function symbols.
- *Fluents* = those relations or functions whose truth values may vary from situation to situation.
 - Denoted by predicate or function symbols taking a situation term as one of their arguments.
 - In a world in which it is possible to paint objects, we might have a functional fluent $colour(x, s)$, that denotes the colour of object x when the world is in situation s .

The Situation Calculus (Continued)

- Actions have *preconditions*: necessary conditions which a world situation must satisfy if the action can be performed in this situation.

- If it is possible for a robot r to pick up an object x in the world situation s then the robot is not holding any object, it is next to x , and x is not heavy:

$$Poss(pickup(r, x), s) \supset [(\forall z) \neg holding(r, z, s)] \wedge \neg heavy(x) \wedge nextto(r, x, s).$$

- Whenever it is possible for a robot to repair an object, then the object must be broken, and there must be glue available:

$$Poss(repair(r, x), s) \supset hasglue(r, s) \wedge broken(x, s).$$

- World dynamics are specified by *effect axioms* which specify the effect of a given action on the truth value of a given fluent.

- The effect on the relational fluent *broken* of a robot dropping an object:

$$fragile(x) \supset broken(x, do(drop(r, x), s)).$$

- A robot repairing an object causes it not to be broken:

$$\neg broken(x, do(repair(r, x), s)).$$

- Painting an object with colour c :

$$colour(x, do(paint(x, c), s)) = c.$$

The Qualification Problem for Actions

- With only the axioms above, we can't prove anything interesting about when an action is possible.

pickup preconditions:

$$Poss(pickup(r, x), s) \supset [(\forall z) \neg holding(r, z, s)] \wedge \neg heavy(x) \wedge nextto(r, x, s).$$

- We can't infer when a *pickup* is possible!
- What about reversing the implication:

$$[(\forall z) \neg holding(r, z, s)] \wedge \neg heavy(x) \wedge nextto(r, x, s) \supset Poss(pickup(r, x), s).$$

- This sentence is *false*! We also need, in the antecedent of the implication:

$$\neg glued_to_floor(x, s) \wedge \neg arms_tied(r, s) \wedge \neg hit_by_10_ton_truck(r, s) \wedge \dots$$

i.e. all the *qualifications* which must be true in order for a *pickup* to be possible!

- Imagine succeeding in enumerating all the qualifications for *pickup*. Would that help? Not really. Suppose all we know is

$$[(\forall z) \neg holding(R, z, S)] \wedge \neg heavy(A) \wedge nextto(R, A, S).$$

We still can't infer $Poss(pickup(R, A), S)$ because we don't know that the qualifications are true!

The Qualification Problem (Continued)

- Intuitively, here's what we want:

- When given only that the “important” qualifications are true:

$$[(\forall z) \neg \text{holding}(R, z, S)] \wedge \neg \text{heavy}(A) \wedge \text{nextto}(R, A, S).$$

and if we *don't know* that any of the “minor” qualifications (*glued_to_floor*(*A*, *S*), *hit_by_10_ton_truck*(*R*, *S*)) is true, infer *Poss*(*pickup*(*R*, *A*), *S*).

- But if we happen to know that one of the “minor” qualifications is true, this will block the inference of *Poss*(*pickup*(*R*, *A*), *S*).
- Historically, this has been seen to be a problem peculiar to reasoning about actions.
- Not so...

$$\text{bird}(x) \wedge \neg \text{penguin}(x) \wedge \neg \text{ostrich}(x) \wedge \neg \text{peking_duck}(x) \wedge \dots \\ \supset \text{fly}(x).$$

Given *bird*(*Tweety*), want intuitively to infer *fly*(*Tweety*).

- Formally, this is the same problem as action qualifications.
 - “Important” qualification: *bird*(*x*).
 - “Minor” qualifications: *penguin*(*x*), *ostrich*(*x*), ...
- This is the classic example of *nonmonotonic reasoning* in AI.

The Qualification Problem (Continued)

- Our approach (for now):

Assume that for each action $A(\vec{x})$, we have an axiom of the form

$$Poss(A(\vec{x}), s) \equiv \Pi_A(\vec{x}, s),$$

where $\Pi_A(\vec{x}, s)$ is a first order formula with free variables $\vec{x}, s$ which does not mention *do*.

We shall call these *action precondition axioms*.

- *Example:*

$$Poss(pickup(r, x), s) \equiv \\ [(\forall z) \neg holding(r, z, s)] \wedge \neg heavy(x) \wedge nextto(r, x, s).$$

- In other words, ignore all the “minor” qualifications.